GE Vernova Rugby Quality

Through the eyes of a Mathematics Graduate

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Purpose and Perspective

I arrived at GE Vernova Rugby with no intention of pretending that a short placement could teach me how a business of this scale should be run. Nor is this report an audit, a set of recommendations, or an attempt to identify what GE Vernova should do differently. I am a recent mathematics graduate entering an industrial Quality environment for the first time. The people I met have accumulated years (and in some cases decades) of experience that cannot be replicated in a few weeks.

My objective was different: **to learn how they think.**

I wanted to enter an unfamiliar team and maximise the amount I could take from it. That meant understanding the processes and terminology, but also watching how people approached problems, how they communicated uncertainty, where they looked first when something went wrong, and how different functions viewed the same problem. Rather than treating each conversation, meeting or factory visit as an isolated experience, I tried to connect them. Something explained to me in an office on Monday might only make sense when I saw it physically on the factory floor several days later.

Mathematics inevitably influenced the way I did this. My instinct was to look for variables, relationships, constraints, classifications and recurring patterns. I found myself asking where information originated, how risk was measured, why one problem received attention before another, how a subjective engineering judgement became a numerical score, and whether a process that appeared efficient locally remained efficient when viewed as part of the wider system.

What follows is therefore neither a definitive description of GE Vernova Rugby nor an assessment of its Quality function. **It is Rugby Quality as I saw it:** a temporary observer's interpretation of the people, systems and decisions I encountered.

For those within GE Vernova, I hope it provides an unusual outside perspective on work that may have become familiar through years of experience. For those reading this outside the company, it serves a second purpose. It is a record of how I approach an unfamiliar environment: observe first, ask questions, test assumptions, connect apparently separate pieces of information and gradually construct a model of the system. The industrial processes described here belong to Quality. The mindset I extracted from them is something I intend to take with me.

The Quality Organisation

Before describing what I learned, it is useful to locate the people in the system. Rugby Quality is not a single activity. During my placement I encountered management, Supplier Quality, Factory Quality Assurance, auditing and Quality-management activities, alongside Engineering and Manufacturing. Brett Bray, Senior Director of Quality Management and Operational Excellence gave me the broadest view of the function and its strategy. Syed Ahmed, Senior Quality Functional Manager, introduced me to Supplier Quality, where Supplier Quality Engineers work with external suppliers to ensure incoming parts and processes meet GE requirements. Ausra, QMS leader and Quality Assurance Engineer, introduced me to auditing, the Quality Management System and the Zero-Defect Framework. On the factory side, Riccardo Panella, a Quality Manager, and the Factory Quality Assurance team showed me how requirements are translated into inspection and acceptance of physical products. Other specialists, including Kinder, Daniel and David, gave me deeper exposure to supplier assurance, APQP, DFMEA and measurement systems.

A First Glance at the System

Within my first few days at Rugby, I realised that the most interesting aspect of Quality was not the number of controls surrounding a product. It was the number of interfaces surrounding it.

A component can move between supplier, engineering, manufacturing, Quality and eventually the customer. Information travels alongside it: drawings, specifications, inspection requirements, risk assessments, lessons learned, Non-Conformance Reports (NCRs) and corrective actions. At every transition, somebody must know what is required, somebody must communicate it, somebody must execute it and somebody must determine whether what happened was acceptable. This creates a deceptively difficult problem.

Across my time at Rugby, I repeatedly encountered issues connected to communication between functions, unclear ownership, incomplete translation of procedures into execution, supplier non-conformance, competing demands on limited resources and reliance on experienced individuals to resolve ambiguity. During one factory discussion, for example, I heard how workers had been instructed to separate particular items without being given sufficiently detailed information about exactly what needed separating. Elsewhere, I heard how less experienced employees could repeatedly depend on experienced colleagues for guidance, consuming the time of the very people most critical to completing complex builds.

Yet it would be misleading to describe Rugby through those problems alone, because much of what I observed was specifically designed to attack them.

I saw Advanced Product Quality Planning (APQP), a structured framework for managing Quality from product concept through production, and Design Failure Mode and Effects Analysis (DFMEA), which systematically anticipates how a design could fail before those failures reach manufacturing. Supplier requirements translated recurring defects into Critical-to-Quality (CTQ) characteristics features whose conformity is particularly important to the product. On the factory floor, I encountered Quality Control Plans (QCPs), GO/NO-GO

gauges, Point of Work Risk Assessments (POWRAs), calibration controls and Safety, Quality, Delivery and Cost (SQDC) performance boards.
The result was not the image of a finished system, nor of a broken one. It was something more interesting: **a system actively trying to learn.** That distinction became important to me.

An NCR identifies a failure, but the NCR itself does not improve anything. An audit identifies a weakness, but the audit itself does not close it. A DFMEA can assign a risk, but a number does not remove that risk. A procedure can describe how work should be performed, but a document cannot guarantee that knowledge has reached the person doing the work.

The real question running though my placement gradually became: **How does an organisation convert information about failure into a change that survives after the immediate problem has disappeared?**

Almost everything I subsequently encountered (from Brett's management philosophy to Ausra's audits, Syed's supplier investigations, DFMEA, factory inspection and measurement-system analysis) could be placed somewhere inside that question.

1 – Brett: Seeing Quality as a System

My introduction to Rugby began with Brett Bray, Senior Director of Quality Management and Operational Excellence. His position gave me an unusually broad entry point into the Quality function: production inspection, Quality engineering, non-conformance visibility, audits, Quality-management initiatives and the allocation of Quality resources.

Brett met me at the entrance, walked me through the site and gave me some of its history before we discussed what I wanted to achieve during this placement. What immediately stood out was the combination of sociability and professionalism with which he operated. Conversations moved easily between people, factory problems, management, risk and long-term strategy. His previous experience commanding submarines also made some of his views on responsibility, escalation and risk particularly interesting to me.

Brett gave me my first useful mental model of Quality.

There is never unlimited time, money or manpower. It is therefore impossible to inspect everything, audit and investigate everything with equal intensity. His language repeatedly returned to concentrating attention on the areas creating the greatest exposure and using the available people, audits and data accordingly. My notes eventually reduced the idea to a simple phrase: **"maximise exposure with minimum resources"**

That immediately felt familiar.

Strip away the industrial terminology and it resembles an optimization problem. There is an objective: reduce Quality exposure, subject to constraints on people, time, information and cost. The interesting question is therefore not simply "what could we inspect?", but "where does another unit of attention create the greatest reduction of risk?"

This was my first indication that mathematics would provide a useful lens without requiring the work itself to be mathematical. The same logic appeared in supplier monitoring. Safety,

Quality, Delivery and Cost (SQDC) could be transformed into weighted measures; while NCR and Cost of poor Quality (CoPQ: the financial impact associated with failures, rework and related Quality issues) could help identify where failures were concentrated. Suppliers could then receive different levels of attention according to performance and risk. The purpose of the measure was not the number. It was **to help decide where to look.**

But Brett's approach also introduced a complication that followed me throughout the placement: an organisation is not an equation. Processes can be redesigned relatively quickly. Dashboards can be built. Data can be reorganised. A new audit can be introduced. Culture moves more slowly. Brett repeatedly distinguished between changing a process and changing the behaviour surrounding that process.

That explains why some of his initiatives were surprisingly informal. The Quality Café, for example, initially seemed almost unrelated to the technical systems I was learning. There was no requirement for somebody to arrive with a presentation or produce a measurable output. People could simply meet and talk. Yet Brett had observed that those conversations naturally migrated towards work: one person would show another a software shortcut, discuss a recurring difficulty or help resolve something that would otherwise have travelled through several formal channels. I found that interesting. A Quality Café is difficult to represent in a spreadsheet, yet it can alter the network through which information moves.

The same philosophy appeared in more structured forms. NCR information was made visible, a Quality SharePoint consolidated resources, visual boards placed information where people would repeatedly encounter it, and the War Room physically brought people around the same problems.

The common variable was not documentation. It was information flow. By the end of my time with Brett, I had started to see Quality less as a department responsible for finding defects and more as an information system trying to decide three things: **Where is the risk? Who needs to know? What happens next?**

The remaining days would show me how difficult each of those questions becomes once people, suppliers and physical machines enter the model.

2 – Ausra: A Process is Only as Strong as its Ownership

If Brett first made me think about the architecture of Quality, Ausra made me think about its discipline.

Ausra is the QMS Leader and Quality Assurance Engineer. Ausra was also one of the people who took the greatest interest in what I would personally take away from Rugby. She was consistently welcoming and made sure that I was learning something from each day, but she also encouraged me to look beyond the individual Quality tools she was showing me. One idea in particular stayed with me: Quality should not be confined to a Quality department. The habit of questioning why something failed, establishing ownership, addressing the underlying cause and then sustaining the improvement is a mindset that can be carried into almost any environment. That changed how I approached the remainder of the placement. I was no longer simply trying to understand Quality terminology; I was increasingly asking which of its principles I could take with me.

My time with Ausra Liaudenskaite exposed me to another side of Quality: The Quality Management System, ISO 9001, internal auditing, competency management and the Zero-Defect Framework. ISO 9001 is the international standard that provides the framework for a Quality Management System (QMS), the processes through which an organisation defines, controls, evaluates and improves how Quality requirements are met. Initially these appeared to be the least mathematical part of my placement: frameworks, clauses, documentation, training records and audits.

They eventually became some of the most interesting. One recurring question was deceptively simple: **Who owns this?**

If a responsibility exists primarily because one employee knows they are supposed to perform it, the process works while that person is present. If they are absent, change role or leave the organisation, part of the process can disappear with them. Ausra's interest in competency frameworks, evidence of training and clearly recorded responsibilities therefore had a purpose beyond documentation: they make organisational knowledge less dependent on memory. This would become much more significant to me later on the factory floor.

Her approach to auditing also changed my initial perception of an audit. I had imagined something closer to an inspection: compare reality with a requirement, identify deviations and record them. What I observed was much more investigative.

Ausra is particularly good at asking repeatedly for information without making the person opposite her feel that the conversation has become an interrogation. She can push, return to a question from another direction and ask for evidence while preserving a surprisingly normal human interaction. That matters because an audit depends on information, and information becomes less useful if the person possessing it becomes defensive. The objective, as she repeatedly presented it to me, was not merely to find something wrong. It was to make sure the problem was **closed permanently.** That distinction seems small until it is treated mathematically.

Suppose a corrective action removes a visible problem at time t. If the underlying mechanism remains unchanged, the probability of recurrence remains. The system has corrected an output, not altered the process generating it. Ausra's emphasis on ownership, objective evidence, follow-up and maturity therefore resembles a feedback system. Detection is only the first signal. Somebody must act on that signal; responsibility must be assigned; implementation must be observed; and the system must later provide evidence that the intervention changed the underlying behaviour.

The maturity framework she showed me made that progression explicit. A control could move from **not implemented**, through **reactive** and **defined**, to **managed**, and eventually **optimised** (where the process becomes measured, data-driven and capable of **sustained** improvement). A process that works once is an event. A process that continues working when the auditor leaves, when the manager changes and when the experienced employee is absent is a **system**.

3 – Syed: Turning Failure into Information

Syed Ahmed, Senior Quality Functional Manager, introduced me to Supplier Quality. Supplier Quality Engineers (SQEs) manage and improve the Quality of parts and materials

coming from external suppliers, working across supplier qualification, process assurance, non-conformance, root-cause analysis and preventive Quality planning.

With Syed, Quality became faster. Problems were no longer primarily clauses or frameworks. They were non-conformances, suppliers, engineering questions, failure mechanisms, presentations, **root causes** and decisions.

What stood out most about Syed was not simply his experience. It was how he communicated it. He could be dynamic and demanding about the technical substance of an investigation while simultaneously making the people around him feel heard. When reviewing an engineering presentation with Giuliano, he repeatedly returned to notation, definitions and clarity before moving step-by-step through what had happened, why it might have happened, why the issue had escaped and what action should follow.

I recognised something immediately. In mathematics, a solution can contain the right intuition and still be a poor solution if its definitions are ambiguous, assumptions are hidden or the argument cannot be followed. Here the consequence is considerably more physical, but the principle survives: if the reasoning cannot be reconstructed, it cannot reliably be challenged.

Supplier Quality made this visible. FMEA (Failure Mode and Effects Analysis) introduced me to an attempt to convert engineering concern into structured risk. **Severity**, **occurrence** and **detection** were assessed and combined into a **Risk Priority Number** (RPN), providing a structured way to prioritise potential failure modes. The objective was not to claim that risk had suddenly become perfectly measurable. It was to impose structure on the decision about what deserved attention first. That distinction matters.

Putting a number on something does not automatically make it objective. The score still depends on definitions, assumptions, scales and human judgement. But a well-designed framework makes those assumptions **visible.** People can challenge why occurrence was rated at one level rather than another, why a detection control is considered effective, or why one failure deserves priority.

The same idea appeared in the 6M Cause Assessment Matrix I later encountered. Potential causes were separated across Man, Machine, Method, Material, Measurement and Environment, with different levels of concern forming what looked almost like a heat map. What interested me was not the colour coding itself, but the decomposition. A statement such as “the component failed” contains almost no useful information.

The Quality process progressively breaks that statement apart:
What failed?
Where?
How severe was it?
How frequently could it occur?
Could we have detected it?
Why did it occur?
Why did it escape?
What must change for it not to happen again?

The problem becomes smaller as the questions become sharper.

4 – From Expertise to Method: Kinder, Daniel and David

By the time I met Kinder, I had encountered several formal methods for managing Quality: supplier scorecards, NCRs, APQP, inspection plans and risk classifications. Kinder gave me a different perspective on the same problem.

He had spent roughly four decades moving between engineering, Quality and operations, across industries ranging from automotive components to compressors before joining GE Vernova. Watching him discuss supplier Quality made it clear that much of what formal systems attempt to capture numerically can also exist, after enough repetition, as professional judgement.

His approach was rigorous without being indiscriminate. When assessing a supplier, he would work through specifications and documentation, question how individual requirements were being achieved and determine whether the supplier actually understood what GE expected. His assessment extended beyond the manufactured component itself to the process and evidence surrounding it.

But what interested me most was that inspection intensity was not constant. A component could have a formal classification indicating its importance, yet Kinder could still devote attention to something apparently less significant if its function made it critical to the wider machine. Similarly, supplier inspection could be concentrated at particular points rather than repeated identically for every unit.

This brought me back to the optimisation problem I had first recognised with Brett. If inspection capacity is finite, checking everything equally is not necessarily the safest strategy. Attention should follow risk, and risk is not synonymous with size, price or formal classification. A relatively small component can become disproportionately important if its failure propagates through the wider system. Kinder's experience appeared to let him make these distinctions rapidly.

That raised a question that would become central to my understanding of Rugby: How much of an expert's decision-making can be encoded into a process, and how much remains inside the expert? It is tempting to treat those as competing approaches: experience on one side and formalisation on the other. I increasingly saw them as complementary. Experience identifies patterns that a generic framework may initially miss. A strong Quality system then tries to capture those lessons so that the organisation does not need to rediscover them through another failure. The challenge is deciding what can be codified without removing the judgement that made the lesson valuable in the first place.

Daniel approached the same question from further upstream. Drawing on extensive automotive Quality experience, he walked me through Advanced Product Quality Planning (APQP): a cross-functional framework intended to manage Quality from concept and design through process development, validation and production. What interested me was that APQP attempts to make prevention systematic rather than dependent on somebody remembering what went wrong last time. Within that framework, Design Failure Mode and Effects Analysis (DFMEA) decomposes a product into systems, subsystems, components and interfaces before asking how each function could fail. Potential effects and causes are then

connected to preventive and detection controls. The exercise therefore converts engineering imagination “what could go wrong?” into a repeatable analytical process.

David extended that logic into measurement and validation. His discussions on DFMEA and Measurement System Analysis (MSA) moved the question one step further: even if the correct characteristic has been identified, how do we know that the system measuring it is reliable? Accuracy, precision, stability, repeatability and gauge variation determine whether the measurement itself is capable of supporting the decision being made.

The three conversations therefore operated at different points of the same chain. Kinder showed me expert judgement applied to a real supplier. Daniel showed how potential failures can be structured before they occur. David showed that even the evidence used to accept or reject something has to be validated. In each case, experience becomes more valuable when part of it can be translated into a method that somebody else can reproduce.

5 – Riccardo: Where the Model Meets the Machine

Riccardo Panella, a Quality Manager working closely with Factory Quality Assurance, gave me the shop-floor counterpart to what I had been learning in Supplier Quality. Factory Quality Assurance focuses on whether the physical product and manufacturing process conform to the required specifications before the build progresses.

Until this point, much of Quality had reached me through presentations, spreadsheets, procedures and conversations. Then I spent considerably more time downstairs. Factory Quality changed the meaning of several terms I had already learned. A Critical-to-Quality characteristic was no longer an entry in a document. It was something somebody physically needed to inspect. A tolerance was no longer an interval around a nominal value. It determined whether a physical component could continue through assembly. A GO/NO-GO check was not simply a binary variable. I saw how a gauge could physically test whether a feature passed one limit while correctly failing the other.

With Riccardo and the Factory Quality Assurance team, I saw inspections taking place before assembly, supplier documentation being checked, components examined visually and physically, and Quality Control Plans connecting those checks to the build process. Different people concentrated on different sections according to expertise and strengths.
Riccardo struck me as particularly engineering oriented. The Quality questions around him were inseparable from the physical machine: fit, assembly, inspection, communication and whether what had been specified could actually be executed.

Andrew, another skilled member of the factory environment, represented another form of expertise. He did not need to turn every observation into a lengthy explanation. His focus appeared much simpler: do the job correctly. When we encountered a component that had been damaged during a manufacturing operation and required rework, his response was practical and immediate: understand the situation, make sure it was controlled and move towards the correct resolution.

These interactions changed my understanding of the relationship between mathematical models and engineering reality. Mathematics often begins by stripping away irrelevant detail. Engineering cannot always afford that luxury. A drawing may specify a dimension. A procedure may define an acceptable range. A control plan may prescribe an inspection. But

the physical component has texture, temperature, accessibility, contamination risk, manufacturing history and interactions with other components.

Experienced inspectors sometimes used sight and touch alongside formal measurements because the product exists in a richer environment than the specification alone can describe. The model is necessary. But the machine is the final test of the model.

6 – Walking the Factory Alone

One of the most useful parts of the placement occurred when nobody was formally teaching me. I was given permission to walk through the factory independently: observe how work was organised, stop at different stations, ask questions and shadow people where appropriate. Without somebody directing my attention, I could test what I had learned against what I noticed myself.

What initially looked like an enormous manufacturing space began to reveal layers of information. Work areas had designated stations containing information about the relevant machine or build, its stage, safety requirements and timeline. Housekeeping expectations were displayed visually. Point of Work Risk Assessments were **signed and dated**. Other boards tracked Quality, engineering, testing and project progress. Measurement equipment was protected, tools organised, assemblies covered against contamination and shafts physically protected.

This mattered because earlier conversations had concentrated heavily on what could go wrong. Walking independently showed me the other half of the system: the large number of ordinary controls that work precisely because nothing happens. A protective sheet preventing contamination is not memorable when it succeeds. A correctly calibrated gauge does not generate an NCR. A clear work instruction does not produce an investigation. Successful preventive Quality is therefore partially invisible.

I also began finding connections without looking for them. Earlier, Syed had shown me Critical-to-Quality supplier material. On the factory floor, I encountered CTQ information again. Earlier, DFMEA had introduced the idea that detection should happen as close as possible to the source of failure. Later, two mechanics explained why testing assemblies in sections was useful: a problem found at an intermediate stage could be isolated before the entire assembly was complete.

Elsewhere, a local whiteboard managed by Ben combined SQDC measures, one-to-one discussions and a section for Continuous Improvement ideas generated by people working directly on the machines. Those ideas could then move into wider discussions and, where appropriate, into drawings and documentation. I found this particularly significant. The information flow was no longer exclusively management → process → worker. It could move in the opposite direction: Worker → observation → improvement idea → process → documentation. **Again, information travelled backwards so that the system could improve forwards.**

7 – The Hidden Variable: Experience

At the beginning of my placement, experience looked like one of Rugby's greatest assets. By the end, I had started to see the other side of the equation. The more knowledge concentrated

inside an individual, the more valuable that individual becomes, but the more vulnerable the system becomes without them.

This became clearest during a factory visit with Ausra. We spoke with an experienced mechanic about training and responsibility within the manufacturing environment. His concern was that less experienced employees repeatedly needed to approach experienced workers for guidance. Those experienced people were themselves required for critical build activities, meaning that every interruption consumed capacity elsewhere.

Viewed individually, each interaction is perfectly rational. Someone does not know how to perform a task. They ask the person who does. But viewed as a system, repeated often enough, the structure creates a feedback loop: Insufficiently embedded knowledge → more questions → more interruptions to experts → reduced expert capacity → greater delays and dependency on the same experts. This observation connected several apparently unrelated parts of my placement. Ausra had spoken about demonstrating competencies and documenting responsibilities so that knowledge did not disappear when somebody was absent.

Kinder, Daniel and David showed me three different ways experienced knowledge can be transformed into structured decisions: supplier judgement, preventive design analysis and measurement validation. Factory inspection showed me how much experienced people can infer from a physical product. The mechanic showed me the operational consequence when too much of that knowledge remains concentrated in those individuals. The conclusion was therefore not that Rugby should replace experience with procedures. That would misunderstand the value of expertise. Nor can every engineering judgement realistically be converted into a checklist. The more interesting challenge is determining which knowledge must remain judgement and which knowledge should become system. That distinction applies far beyond Quality.

An organisation wants experienced people because they can recognise situations that formal rules cannot fully anticipate. But it also wants their repeated lessons to survive them. Every time a useful piece of tacit knowledge becomes a robust training method, control, drawing, procedure or design rule, some part of individual experience becomes organisational capability. That, increasingly, was how I understood many of the initiatives I had encountered at Rugby. APQP, DFMEA, audit frameworks, CTQ communications, Quality Control Plans, training matrices, visual management and lessons learned all perform different functions. But at another level, each is attempting to do something similar: make useful knowledge reproducible.

8 – Measuring the Measurement

My final discussions with David added one last complication: even if a process is defined, a specification is correct and somebody performs the required measurement, there remains another question: How reliable is the measurement itself?

Measurement System Analysis (MSA) introduced variation not simply as something present in the manufactured product, but as something that can be introduced by the system used to measure it. The discussions covered accuracy, precision, stability, repeatability and the variation attributable to gauges and measurement processes. The purpose was to understand

whether a measurement system was sufficiently capable to support the decisions being made from its output.

For me, this completed an interesting circle. At the beginning of the placement, numbers had appeared to offer certainty. Supplier scores, audit scores, RPNs, NCR rates and other metrics converted complex situations into values that could be compared. By the end, I had become more cautious. A number is the end of a chain.

Before trusting it, I need to ask:
Where did the data come from?
What exactly was measured?
How was it measured?
How much variation does the measurement system introduce?
How were categories or weights chosen?
What information disappeared when reality was compressed into the metric?

This does not make quantitative analysis less valuable. It makes it more valuable when used correctly. The lesson I took from Quality was therefore not “measure everything.” It was: Understand the measurement before trusting the measure. And for a mathematics graduate, that may have been one of the most useful lessons of the placement.

9 – From Detection to Prediction

On my first day, Brett discussed an ambition that immediately caught my attention: using data and potentially AI not merely to report Quality problems, but eventually to anticipate where they are likely to occur.

At the time, prediction sounded like a separate, future-facing topic. By the end of the placement, I saw it differently. Prediction is difficult without prevention. Prevention is difficult without understanding. Understanding is difficult without reliable detection. Reliable detection is difficult without good measurement. And none of those stages work sustainably if information remains fragmented or responsibility is unclear.

The progression I observed can therefore be represented simply: Detect → Measure → Understand → Prevent → Sustain → Predict

But the arrows are more important than the words. Each transition requires information to move: from the factory into an NCR, from an NCR into root-cause analysis, from root cause into supplier or process action, from an action into a revised control, from experience into documentation, and eventually from accumulated historical information into an ability to recognise risk before failure occurs. That is why my view of Quality changed so substantially. I had arrived associating Quality primarily with checking whether something was correct. I left seeing it much more as the management of information, uncertainty and learning around a physical system.

Conclusion: What I Took from Rugby

I did not leave Rugby as a Quality engineer, and that was never the objective. I left with a different way of looking at systems.

Brett made me think about constrained resources, information flow and the relationship between process and culture. Ausra made me think about ownership, evidence and whether a corrective action survives after attention moves elsewhere. Syed showed me how complex failures can be decomposed into progressively sharper questions. Kinder showed me the value of accumulated supplier judgement; Daniel how potential failure can be structured before it occurs; and David how the measurement itself must be challenged before its output is trusted. Riccardo, Andrew and Factory Quality Assurance showed me where those formal systems encounter physical reality.

What initially looked like a collection of unrelated Quality tools gradually became connected. An NCR, an audit, a DFMEA, a GO/NO-GO gauge, a training matrix, an SQDC board and a conversation between two experienced engineers look very different. Yet each is, in some form, an attempt to reduce uncertainty and improve the next decision. That is where mathematics ultimately fitted into my placement. Not through complicated equations. Through the instinct to decompose a system, identify its variables and constraints, question the reliability of its measurements, look for dependencies, trace causes rather than symptoms and ask whether a solution remains valid when viewed beyond the immediate problem.

The most important insight was also the least mathematical. A theoretically optimal process is of little value if people cannot understand it, use it or sustain it. Equally, an organisation built entirely on excellent individuals remains vulnerable if their knowledge cannot travel beyond them. Quality therefore seemed to exist somewhere between the two: process and judgement, measurement and experience, prevention and response, people and system.

I arrived at Rugby trying to understand what the Quality team did. I left more interested in how they think. That is the part I intend to carry forward.